

Evaluating Bempedoic Acid for Non-alcoholic Fatty Liver Disease: A Review of Preclinical and Clinical Research.

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Non-alcoholic fatty liver disease (NAFLD) is the most prevalent liver disorder globally, characterized by fat accumulation in liver cells, which can progress to inflammation, fibrosis, and cirrhosis. The disease predominantly affects individuals with obesity and high body mass index (BMI). It is a globally prevalent condition, with variations in incidence across different regions. The pathophysiology of NAFLD involves insulin resistance, metabolic disturbances, and genetic and gut microbial factors. Current treatments primarily focus on lifestyle modifications and a limited range of pharmacological options. Bempedoic acid (BA), a novel cholesterol-lowering agent, targets adenosine triphosphate (ATP)-citrate lyase to reduce low-density lipoprotein (LDL) cholesterol and has shown potential in managing NAFLD by decreasing liver fat accumulation and improving lipid profiles. BA's unique mechanism offers a promising add-on therapy, particularly for the patient's intolerant to statins. Despite its potential, comprehensive clinical and preclinical studies are needed to further elucidate its efficacy and safety compared to other NAFLD treatments. Future research should focus on comparing BA with existing and emerging therapies to optimize its role in NAFLD management and enhance patient outcomes.

The small extracellular vesicle-mediated intercellular transformation of CXCR1^{Low} to CXCR1^{High} tumour cells promotes the progression of head and neck squamous cell carcinoma.

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The heterogeneity of tumour cells enables cancers to dynamically adapt to microenvironmental stresses during progression. However, the mechanism underlying the transformation and intercellular communication between heterogeneous tumour cells has remained elusive. Here, we report a "contagion model" that mediates intercellular transformation between heterogeneous tumour cells which facilitates tumour progression. Initially identifying heterogeneous expression of CXCR1, a receptor for interleukin-8, in head and neck squamous cell carcinoma (HNSCC) tumour cells, we found that CXCR1^{High} tumour cells had higher abilities for migration and invasion. Following interleukin-8-mediated activation, CXCR1^{High} cells transformed CXCR1^{Low} cells into CXCR1^{High} cells through the secretion of small extracellular vesicles (sEVs), which increased the proportion of CXCR1^{High} cells and facilitated tumour progression. Mechanistically, we demonstrate that sEVs derived from interleukin-8-activated CXCR1^{High} cells contain high levels of ATP citrate lyase (ACLY), which acetylates NF- κ B p65 and facilitates its nuclear translocation to transcribe CXCR1 in CXCR1^{Low} cells. That process could be inhibited by Bempedoic acid, an FDA-approved ACLY-targeted drug. Taken together, our study reveals an sEV-mediated transformation of CXCR1^{Low} to CXCR1^{High} cells that promotes HNSCC progression. This provides a new paradigm to explain the dynamic changes of heterogeneous tumour cells, and identifies Bempedoic acid as a potential drug for HNSCC treatment.

Mechanism of action and therapeutic use of bempedoic acid in atherosclerosis and metabolic syndrome.

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Bempedoic acid is a new cholesterol-lowering drug, which has recently received US FDA and EMA approval. This drug targets lipid and glucose metabolism as well as inflammation via downregulation of ATP-citrate lyase and upregulation of AMP-activated protein kinase (AMPK). The primary effect is the reduction of cholesterol synthesis in the liver and its administration is generally not associated to unwanted muscle effects. Suppression of hepatic fatty acid synthesis leads to decreased triglycerides and, possibly, improved non-alcoholic fatty liver disease. Bempedoic acid may decrease gluconeogenesis leading to improved insulin sensitivity, glucose metabolism, and metabolic syndrome. The anti-inflammatory action of bempedoic acid is mainly achieved via activation of AMPK pathway in the immune cells, leading to decreased plasma levels of C-reactive protein. Effects of bempedoic acid on atherosclerotic cardiovascular disease, type 2 diabetes and chronic liver disease have been assessed in randomized clinical trials but require further confirmation. Safety clinical trials in phase III indicate that bempedoic acid administration is generally well-tolerated in combination with statins, ezetimibe, or proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors to achieve low-density lipoprotein cholesterol targets. The aim of this narrative review on bempedoic acid is to explore the underlying mechanisms of action and potential clinical targets, present existing evidence from clinical trials, and describe practical management of patients.

Bempedoic acid as a PPAR α activator: new perspectives for hepatic steatosis treatment in a female rat experimental model.

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In its initial stages, nonalcoholic fatty liver disease presents hypertriglyceridemia and accumulation of lipids in the liver (hepatic steatosis). Bempedoic acid is an ATP:citrate lyase inhibitor that promotes a dual inhibition of the synthesis of cholesterol and fatty acids. However, its effect in the prevention / treatment of hepatic steatosis and hypertriglyceridemia has not been investigated. The aim of our work has been to elucidate whether bempedoic acid, through a mechanism other than ATP:citrate lyase inhibition, reverses these metabolic alterations. The study was carried out in female Sprague-Dawley rats fed, for three months, with a high fat diet supplemented with fructose (10% w/v) in drinking water. During the last month, bempedoic acid (30mg/kg/day) was administered to a group of animals. Zoometric and plasmatic parameters were analyzed, gene and protein expression analysis were performed in liver samples and PPAR-PPRE binding activity was determined. Our interventional model developed hepatic steatosis and hypertriglyceridemia. Despite an increase in total caloric intake, there was no increase in body weight of the animals. The administration of bempedoic acid significantly reduced hepatic steatosis and promoted a marked hepatocyte hypertrophy. There was a 66% increase in the liver weight of the animals treated with the drug that was not accompanied by modifications in the markers of inflammation, oxidative stress, or endoplasmic reticulum stress. Bempedoic acid activated the peroxisome proliferator activated nuclear receptor (PPAR α) and its target genes. Bempedoic acid could be an effective therapy for the treatment of fatty liver and associated cardiovascular risk. Bempedoic acid has other mechanisms of action besides the inhibition of ATP: citrate lyase, such as the activation of PPAR α , which could explain the reduction in hepatic steatosis and the increase in liver weight observed in animals treated with the drug.

ATP citrate lyase inhibitor Bempedoic Acid alleviate long term HFD induced NASH through improvement in glycemic control, reduction of hepatic triglycerides & total cholesterol, modulation of inflammatory & fibrotic genes and improvement in NAS score.

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Non-alcoholic fatty liver disease (NAFLD) and Non-alcoholic steatohepatitis (NASH) are chronic liver disorders, the prevalence of which is increasing worldwide. Long term High Fat Diet (HFD) induced NASH animal models closely mimic the characteristics of human NASH and hence used by investigators as a model system for studying the mechanism of action of new drugs. Bempedoic acid (ETC-1002), a ATP citrate lyase (ACLY) inhibitor that lowers the LDL cholesterol was recently approved by US FDA for the treatment of heterozygous familial hypercholesterolemia (HeFH) and established atherosclerotic cardiovascular disease (ASCVD). ACLY is one of the genes modulated in NASH patients and hence we studied the effect of ACLY inhibitor Bempedoic acid in long term HFD induced NASH animal model to understand the pharmacological benefits and the associated mechanism of action of this newly approved drug in NASH. Mice fed with 60% Kcal High Fat Diet for 32 weeks were used for the study and the animals were given Bempedoic acid for 5 weeks at doses of 10

mg

kg-1, po, qd, and 30

mg

kg-1, po, qd. Bempedoic acid treatment resulted in inhibition of body weight gain and improved the glycemic control. Bempedoic acid treated group showed statistically significant reduction in plasma ALT, AST, hepatic triglycerides (TG) and total cholesterol (TC), along with statistically significant reduction in steatosis score by histological analysis. Hepatic gene expression analysis showed significant reduction in inflammatory and fibrotic genes such as Mcp-1/Ccl2, Timp-1 & Coll α 1. Histological analysis showed significant improvement in NAS score. Overall, Bempedoic acid alleviated HFD induced Non-Alcoholic Steatohepatitis through inhibition of body weight gain, improvement in glycemic control, reduction of hepatic triglycerides & total cholesterol, modulation of inflammatory & fibrotic genes, and improvement in NAS score. Hence, Bempedoic acid can be a potential therapeutic option for metabolic syndrome and NASH.

Bempedoic Acid in the Treatment of Patients with Dyslipidemias and Statin Intolerance.

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An elevated plasma low-density lipoprotein cholesterol (LDL-C) level is a well-established atherosclerotic cardiovascular disease (ACSVD) risk factor. Randomized studies with statins (alone or in combination with other lipid-lowering drugs) have demonstrated their clinical efficacy in lowering LDL-C. Several classes of new, non-statin agents have been successfully studied and used (e.g., ezetimibe and inhibitors of proprotein convertase subtilisin/kexin type 9 [i-PSCK9]). However, many high ACSVD risk patients remain

at a high residual cardiovascular risk, with at least 10% being statin intolerant. Bempedoic acid (ETC-1002) is a new inhibitor of cholesterol synthesis that targets ATP citrate lyase (ACL). Importantly, ETC-1002 is only converted into an active form in the liver and is free of muscle side effects. **Area Covered:** Mechanism of action of ETC-1002, clinical pharmacology, completed clinical studies with bempedoic acid, lipid-lowering efficacy/safety issues, and recent meta-analyses of trials with ETC-1002. **Expert Opinion:** ETC-1002 has been extensively studied in phase I-III clinical studies in over 4000 individuals from different patient populations (statin intolerance, familial hypercholesterolemia, and high ACSVD risk patients), ETC-1002 has been demonstrated to have moderate cholesterol-lowering efficacy and a good safety profile at a dose of 180 mg/day as a monotherapy and in combination with statins and ezetimibe. The ongoing study CLEAR Outcomes, with composite cardiovascular endpoints, will elucidate the role of bempedoic acid in the management of high ACSVD risk and statin-intolerant patients with hypercholesterolemia. Long-term safety data on bempedoic acid are needed to fully establish this agent in evidence-informed guidelines for managing of patients with dyslipidemias.

New, Novel Lipid-Lowering Agents for Reducing Cardiovascular Risk: Beyond Statins.

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Statins are the cornerstone of the prevention and treatment of atherosclerotic cardiovascular disease (ASCVD). However, even under optimal statin therapy, a significant residual ASCVD risk remains. Therefore, there has been an unmet clinical need for novel lipid-lowering agents that can target low-density lipoprotein cholesterol (LDL-C) and other atherogenic particles. During the past decade, several drugs have been developed for the treatment of dyslipidemia. Inclisiran, a small interfering RNA that targets proprotein convertase subtilisin/kexin type 9 (PCSK9), shows comparable effects to that of PCSK9 monoclonal antibodies. Bempedoic acid, an ATP citrate lyase inhibitor, is a valuable treatment option for the patients with statin intolerance. Pemaifibrate, the first selective peroxisome proliferator-activated receptor alpha modulator, showed a favorable benefit-risk balance in phase 2 trial, but the large clinical phase 3 trial (PROMINENT) was recently stopped for futility based on a late interim analysis. High dose icosapent ethyl, a modified eicosapentaenoic acid preparation, shows cardiovascular benefits. Evinacumab, an angiopoietin-like 3 (ANGPTL3) monoclonal antibody, reduces plasma LDL-C levels in patients with refractory hypercholesterolemia. Novel antisense oligonucleotides targeting apolipoprotein C3 (apoC3), ANGPTL3, and lipoprotein(a) have significantly attenuated the levels of their target molecules with beneficial effects on associated dyslipidemias. Apolipoprotein A1 (apoA1) is considered as a potential treatment to exploit the athero-protective effects of high-density lipoprotein cholesterol (HDL-C), but solid clinical evidence is necessary. In this review, we discuss the mode of action and clinical outcomes of these novel lipid-lowering agents beyond statins.

Bempedoic acid: Review of a novel therapy in lipid management.

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An update on clinical development of a first-in-class oral medication for adjunctive cholesterol lowering in high-risk patients with persistent elevation of low-density lipoprotein cholesterol (LDL-C) despite statin therapy is reviewed. Despite the proven efficacy of statin therapy, many patients cannot reach LDL-C goals with statins alone, largely due to inadequate response or intolerance. Nonstatin treatment options to reduce LDL-C include ezetimibe, bile acid sequestrants, and proprotein convertase subtilisin/kexin 9

(PCSK9) inhibitors; however their use has been limited by modest clinical benefit or high treatment costs. Novel nonstatin treatments are in development to further address the needs of this population. Bempedoic acid is a first-in-class oral adenosine triphosphate (ATP) citrate lyase inhibitor being evaluated as an additional treatment option for high-risk patients requiring further reduction in LDL-C. Bempedoic acid has been evaluated in multiple phase 2 and phase 3 trials as monotherapy or for use in combination with ezetimibe and/or statin therapy. Treatment with bempedoic acid has been demonstrated to result in significant reductions in LDL-C and several other cardiovascular risk markers without the myalgia associated with statin therapy. Bempedoic acid, used alone or with ezetimibe in a fixed-dose combination formulation, may be an effective alternative to current guideline-recommended nonstatin therapies in patients who do not attain adequate LDL-C lowering with maximally tolerated statin therapy and in statin-intolerant patients at risk for atherosclerotic cardiovascular disease.

Bempedoic Acid and Cardiovascular Outcomes in Statin-Intolerant Patients.

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Bempedoic acid, an ATP citrate lyase inhibitor, reduces low-density lipoprotein (LDL) cholesterol levels and is associated with a low incidence of muscle-related adverse events; its effects on cardiovascular outcomes remain uncertain. We conducted a double-blind, randomized, placebo-controlled trial involving patients who were unable or unwilling to take statins owing to unacceptable adverse effects ("statin-intolerant" patients) and had, or were at high risk for, cardiovascular disease. The patients were assigned to receive oral bempedoic acid, 180 mg daily, or placebo. The primary end point was a four-component composite of major adverse cardiovascular events, defined as death from cardiovascular causes, nonfatal myocardial infarction, nonfatal stroke, or coronary revascularization. A total of 13,970 patients underwent randomization; 6992 were assigned to the bempedoic acid group and 6978 to the placebo group. The median duration of follow-up was 40.6 months. The mean LDL cholesterol level at baseline was 139.0 mg per deciliter in both groups, and after 6 months, the reduction in the level was greater with bempedoic acid than with placebo by 29.2 mg per deciliter; the observed difference in the percent reductions was 21.1 percentage points in favor of bempedoic acid. The incidence of a primary end-point event was significantly lower with bempedoic acid than with placebo (819 patients [11.7%] vs. 927 [13.3%]; hazard ratio, 0.87; 95% confidence interval [CI], 0.79 to 0.96; $P = 0.004$), as were the incidences of a composite of death from cardiovascular causes, nonfatal stroke, or nonfatal myocardial infarction (575 [8.2%] vs. 663 [9.5%]; hazard ratio, 0.85; 95% CI, 0.76 to 0.96; $P = 0.006$); fatal or nonfatal myocardial infarction (261 [3.7%] vs. 334 [4.8%]; hazard ratio, 0.77; 95% CI, 0.66 to 0.91; $P = 0.002$); and coronary revascularization (435 [6.2%] vs. 529 [7.6%]; hazard ratio, 0.81; 95% CI, 0.72 to 0.92; $P = 0.001$). Bempedoic acid had no significant effects on fatal or nonfatal stroke, death from cardiovascular causes, and death from any cause. The incidences of gout and cholelithiasis were higher with bempedoic acid than with placebo (3.1% vs. 2.1% and 2.2% vs. 1.2%, respectively), as were the incidences of small increases in serum creatinine, uric acid, and hepatic-enzyme levels. Among statin-intolerant patients, treatment with bempedoic acid was associated with a lower risk of major adverse cardiovascular events (death from cardiovascular causes, nonfatal myocardial infarction, nonfatal stroke, or coronary revascularization). (Funded by Esperion Therapeutics; CLEAR Outcomes ClinicalTrials.gov number, NCT02993406.).

Bempedoic acid: a promising novel agent for LDL-C lowering.

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Bempedoic acid (ETC-1002) is a novel, first-in-class, oral, small molecule that inhibits cholesterol biosynthesis in the same pathway as statins, thereby lowering low-density lipoprotein cholesterol (LDL-C) by upregulating LDL receptors. Preclinical and completed Phase II and III clinical trials have demonstrated promising results regarding its safety and efficacy across a variety of patient characteristics including statin intolerance and on a background of lipid-lowering therapy. Bempedoic acid is currently being evaluated in a cardiovascular outcomes trial to evaluate its effect on major cardiovascular events in patients with or at high risk for cardiovascular disease and with statin intolerance. In this review, we will discuss the history and development of bempedoic acid, relevant clinical trials, and its potential role as a lipid-lowering medication in the context of other currently available lipid-lowering therapies.
